

# VIVEK CHAUDHARY

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## Summary

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Software Engineering student with hands-on experience in full-stack development, AI-driven applications, and backend systems using C++, Python, JavaScript, TypeScript, React.js, Next.js, FastAPI, and Node.js. Strong foundation in Data Structures and Algorithms, OOP, DBMS, OS, Computer Networks, and SQL.

## Education

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**Harcourt Butler Technical University, Kanpur**

Bachelor of Technology in Information Technology (CGPA: 7.24)

September 2023 - Present

Kanpur, Uttar Pradesh

## Skills

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**Programming Languages:** C++, Python, JavaScript, TypeScript

**Web Development:** React.js, Next.js, Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, TailwindCSS

**AI / Machine Learning:** LLM Applications, Prompt Engineering, RAG, Groq API, Pydantic, Resume Parsing

**CS Fundamentals:** Data Structures and Algorithms, OOP, DBMS, OS, Computer Networks, SQL

**Tools:** Git, GitHub, Vite, CMake, uv

## Projects

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**HireMeAI – AI-Powered Portfolio & Career Assistant**

[GitHub](#) — [Live](#)

*React 19, TypeScript, Vite, FastAPI, Groq, Pydantic, TailwindCSS*

- Built a full-stack AI portfolio assistant enabling recruiters to query skills, projects, education, and experience using natural language, with responses grounded in resume data.
- Developed FastAPI APIs integrating Groq LLMs for PDF resume parsing and grounded answer generation using **pdf** and **Pydantic** for extraction and validation.
- Implemented a responsive React/TypeScript chat interface with conversation history, new/switch/delete/retry flows, theme support, resume download, and graceful AI/backend error handling.

**DPI Engine – Deep Packet Inspection System**

[GitHub](#)

*C++, CMake, PCAP, TCP/IP, TLS, Multi-threading*

- Developed a C++ Deep Packet Inspection engine for PCAP analysis, parsing Ethernet/IP/TCP headers and tracking network flows using five-tuples.
- Implemented TLS Server Name Indication (SNI) extraction and application classification to identify domains from HTTPS Client Hello traffic.
- Built rule-based traffic filtering using IP, application, and domain rules, with packet forwarding/dropping and traffic statistics.
- Designed a multi-threaded processing pipeline using load-balancer and fast-path worker threads, consistent hashing, and thread-safe queues for parallel packet processing.

## Achievements

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- Secured **10th position among 100+ teams** in Smart India Hackathon 2025 Internal Hackathon at HBTU, Kanpur.
- Active competitive programmer on LeetCode, solving DSA problems consistently.
- Solved 100+ data structures and algorithms problems across various platforms.
- Completed **Dynamic Programming Bootcamp, AlgoUniversity** (May 2026) – [Verify](#)